Initializing the Library

Once an instance of the library has been created, and sometimes even when we don't need to create an instance for the library, we will then usually need to initialize the library somehow. Usually this function is placed inside the `setup()` function, like we did with the LCD example, by using the following line:

```
lcd.begin(16, 2);
```

In this line we use a function called `begin()` that is a part of the LiquidCrystal library linked to the instance that we created called `lcd`. The instance name and function name are separated by a period. In this case, two parameters are passed to this function, which tell the library that we have a display that is 16 characters wide and 2 characters tall.

Every library is a little different and beyond this point what can be done with them begins to significantly diverge. So let's start with our first hardware library that includes a more thorough discussion of the LiquidCrystal library.

LiquidCrystal library

Earlier in this book we looked at one way to see the information coming out of the Arduino using both the Serial Monitor and code that uses the Serial library, which we will revisit in the next chapter. We can also use any of a number of different displays to show visual information and even create a simple user interface to provide visual feedback. These displays could include the simple monochrome character LCD that we used in the last chapter, a color graphical LCD, seven-segment and alphanumeric LEDs, and even text overlays for composite video or e-ink displays. Since we can't cover all of these here, we are going to go back to our character LCD that we used in the last project and look at some of the other things that the library will allow us to do with code examples, which will include displaying text, working with symbols and custom characters, and even creating very basic animations.

Obviously, to use the LiquidCrystal library we need an LCD. The schematic and illustration in Figures 9-1 and 9-2 show a simpler circuit than the one from last chapter that will be used for the next few example sketches. LCDs like these use the HD44780 driver and have a parallel interface with up to eight data bus pins, but we are only going to use four marked DB4-7. In addition, we need to connect the RS or register select pin to tell the display whether to display data or perform a command and the EN or enable pin to let the display know that data is waiting for it. The RW or read/write pin is simply tied to ground to keep the display in a read state and the V0 pin controls the contrast of the display and can be connected to the middle pin of a trimpot or to a resistor. To keep things simple, we have connected the positive or anode side of the backlight LED marked LED+ to a 220 ohm resistor that is then connected to +5v.